

Assignment 3

Computer Programming with C

MCA102

Topic: Arrays and Structure

- 1) **Write a program to store marks for n number of students in an array and print their marks.**

Input :-

```
#include <stdio.h>

int main() {

    int n;

    printf("Enter the number of students: ");

    scanf("%d", &n);

    int marks[n];

    printf("Enter marks for %d students:\n", n);

    for (int i = 0; i < n; i++) {

        printf("Student %d: ", i + 1);

        scanf("%d", &marks[i]);

    }

    printf("\nMarks of all students:\n");

    for (int i = 0; i < n; i++) {

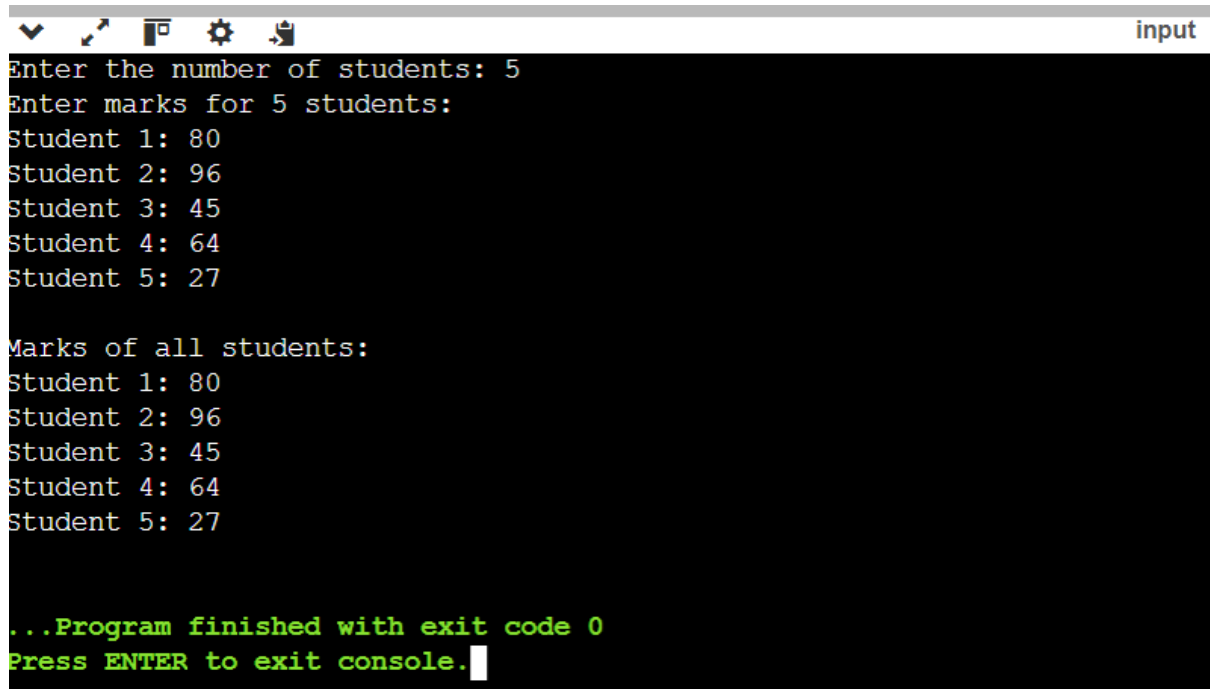
        printf("Student %d: %d\n", i + 1, marks[i]);

    }

    return 0;
```

}

Output :-



```
input
Enter the number of students: 5
Enter marks for 5 students:
Student 1: 80
Student 2: 96
Student 3: 45
Student 4: 64
Student 5: 27

Marks of all students:
Student 1: 80
Student 2: 96
Student 3: 45
Student 4: 64
Student 5: 27

...Program finished with exit code 0
Press ENTER to exit console.
```

- 2) **Write a program that stores the marks of the subject Mathematics and English of n number of students in an array and then prints their total marks.**

Input :-

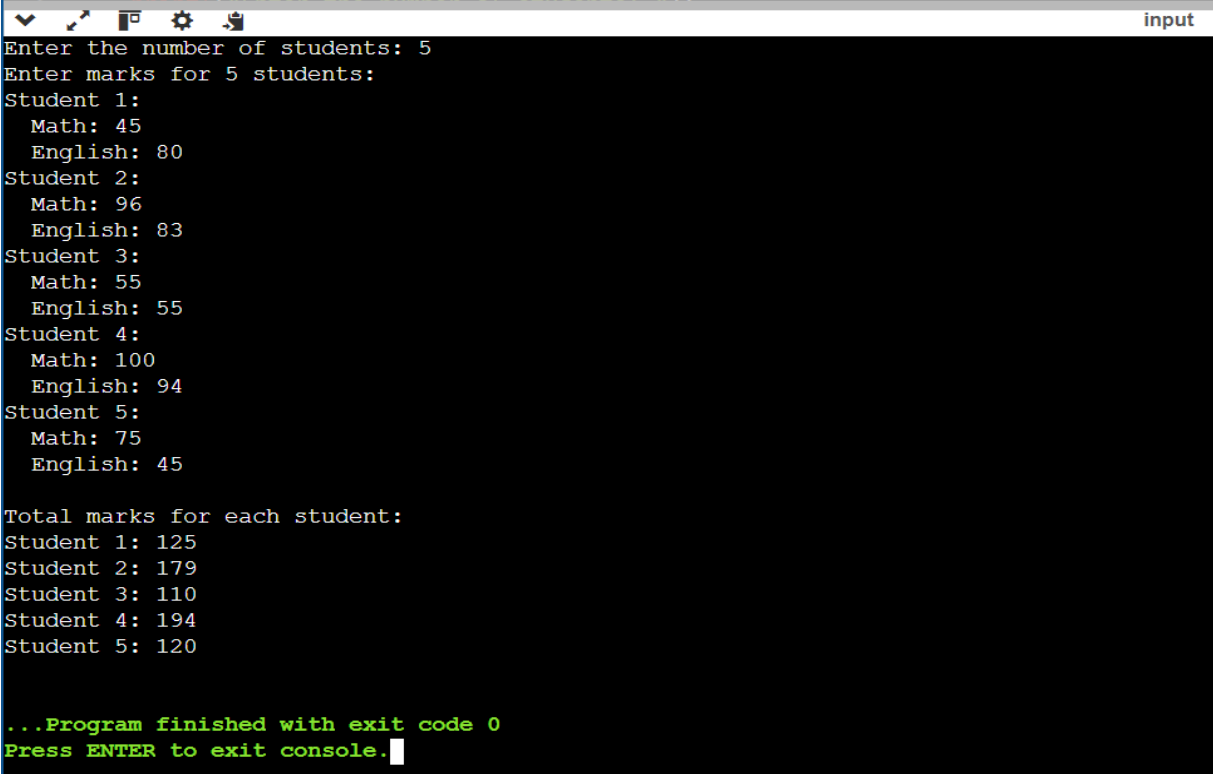
```
#include <stdio.h>
int main() {
    int n;
    printf("Enter the number of students: ");
    scanf("%d", &n);
    int mathMarks[n], englishMarks[n];
    printf("Enter marks for %d students:\n", n);
    for (int i = 0; i < n; i++) {
        printf("Student %d:\n", i + 1);
        printf("  Math: ");
        scanf("%d", &mathMarks[i]);
        printf("  English: ");
        scanf("%d", &englishMarks[i]);
    }
}
```

```

printf("\nTotal marks for each student:\n");
for (int i = 0; i < n; i++) {
    int totalMarks = mathMarks[i] + englishMarks[i];
    printf("Student %d: %d\n", i + 1, totalMarks);
}
return 0;
}

```

Output :-



```

input
Enter the number of students: 5
Enter marks for 5 students:
Student 1:
  Math: 45
  English: 80
Student 2:
  Math: 96
  English: 83
Student 3:
  Math: 55
  English: 55
Student 4:
  Math: 100
  English: 94
Student 5:
  Math: 75
  English: 45

Total marks for each student:
Student 1: 125
Student 2: 179
Student 3: 110
Student 4: 194
Student 5: 120

...Program finished with exit code 0
Press ENTER to exit console.

```

- 3) **Write a program to insert an element in an array in a particular position.**

Input :-

```

#include <stdio.h>
int main() {
    int n, pos, element;
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);

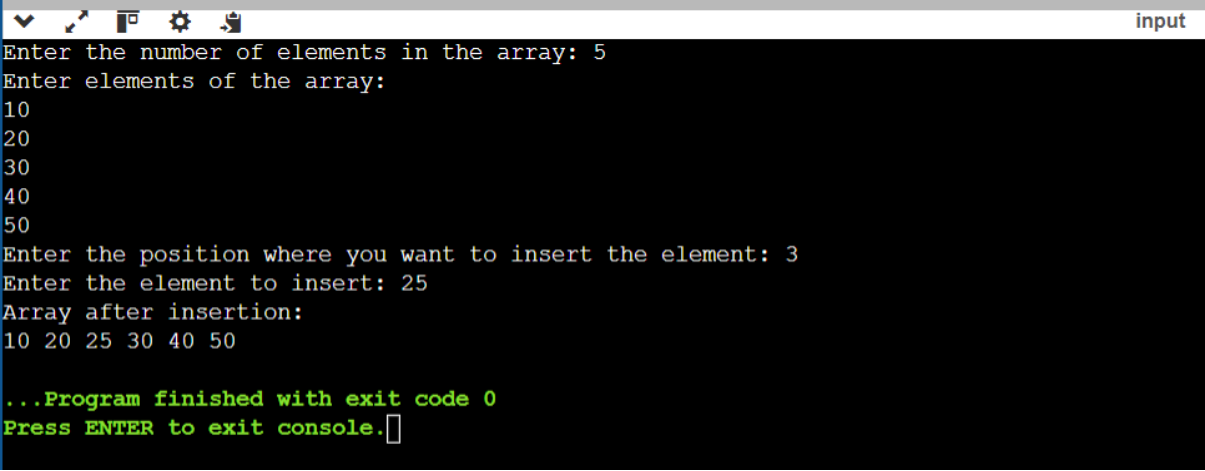
```

```

int arr[n];
printf("Enter elements of the array:\n");
for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}
printf("Enter the position where you want to insert the element: ");
scanf("%d", &pos);
if (pos < 1 || pos > n + 1) {
    printf("Invalid position!\n");
    return 1;
}
printf("Enter the element to insert: ");
scanf("%d", &element);
for (int i = n - 1; i >= pos - 1; i--) {
    arr[i + 1] = arr[i];
}
arr[pos - 1] = element;
printf("Array after insertion:\n");
for (int i = 0; i <= n; i++) {
    printf("%d ", arr[i]);
}
return 0;
}

```

Output :-



The screenshot shows a terminal window with a title bar containing standard icons and the word "input". The terminal output is as follows:

```

Enter the number of elements in the array: 5
Enter elements of the array:
10
20
30
40
50
Enter the position where you want to insert the element: 3
Enter the element to insert: 25
Array after insertion:
10 20 25 30 40 50

...Program finished with exit code 0
Press ENTER to exit console.

```

- 4) **Write a program to delete an element from a particular position of an array.**

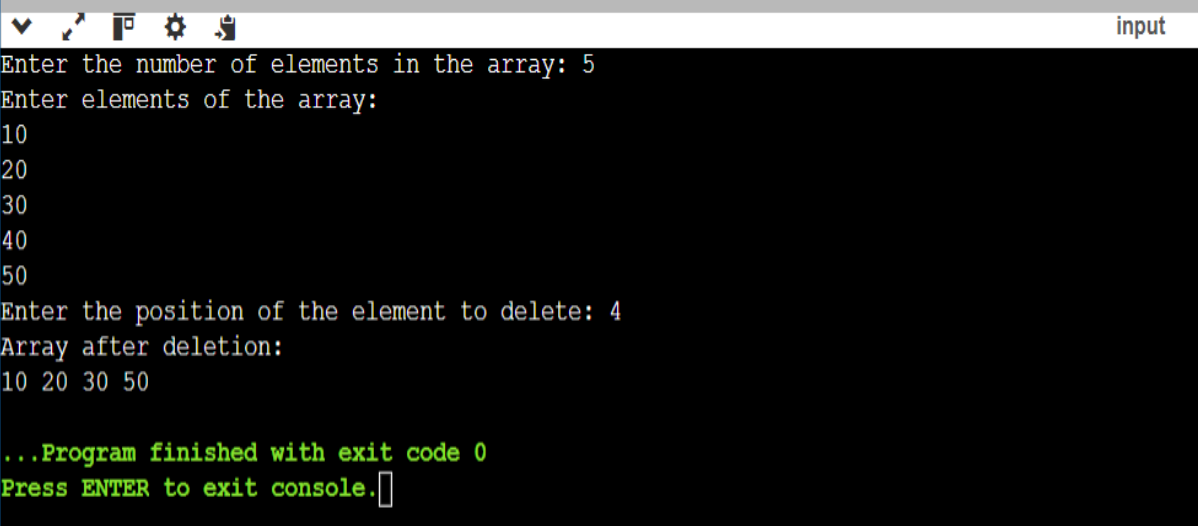
Input :-

```

#include <stdio.h>
int main() {
    int n, pos;
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements of the array:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the position of the element to delete: ");
    scanf("%d", &pos);
    if (pos < 1 || pos > n) {
        printf("Invalid position!\n");
        return 1;
    }
    for (int i = pos - 1; i < n - 1; i++) {
        arr[i] = arr[i + 1];
    }
    printf("Array after deletion:\n");
    for (int i = 0; i < n - 1; i++) {
        printf("%d ", arr[i]);
    }
    return 0;
}

```

Output :-



```

input
Enter the number of elements in the array: 5
Enter elements of the array:
10
20
30
40
50
Enter the position of the element to delete: 4
Array after deletion:
10 20 30 50

...Program finished with exit code 0
Press ENTER to exit console.

```

- 5) **Write a program to convert a decimal number taken as input from a user to the corresponding binary number and store the result in an array.**

Input :-

```
#include <stdio.h>

int main() {

    int decimalNum, binaryNum[32], i = 0;

    printf("Enter a decimal number: ");

    scanf("%d", &decimalNum);

    while (decimalNum > 0) {

        binaryNum[i] = decimalNum % 2;

        decimalNum /= 2;

        i++;

    }

    for (int j = i - 1; j >= 0; j--) {

        printf("%d", binaryNum[j]);

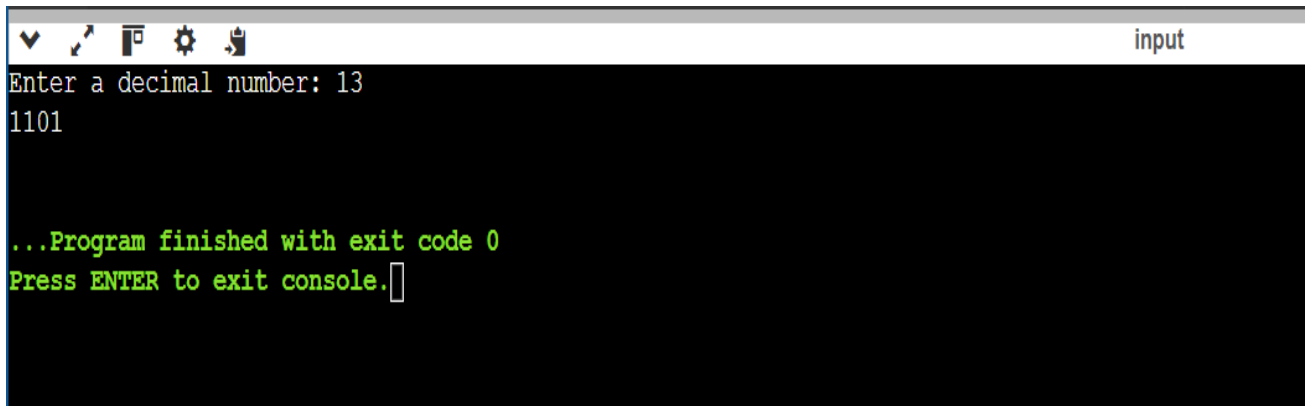
    }

    printf("\n");

    return 0;

}
```

Output :-



```
input
Enter a decimal number: 13
1101

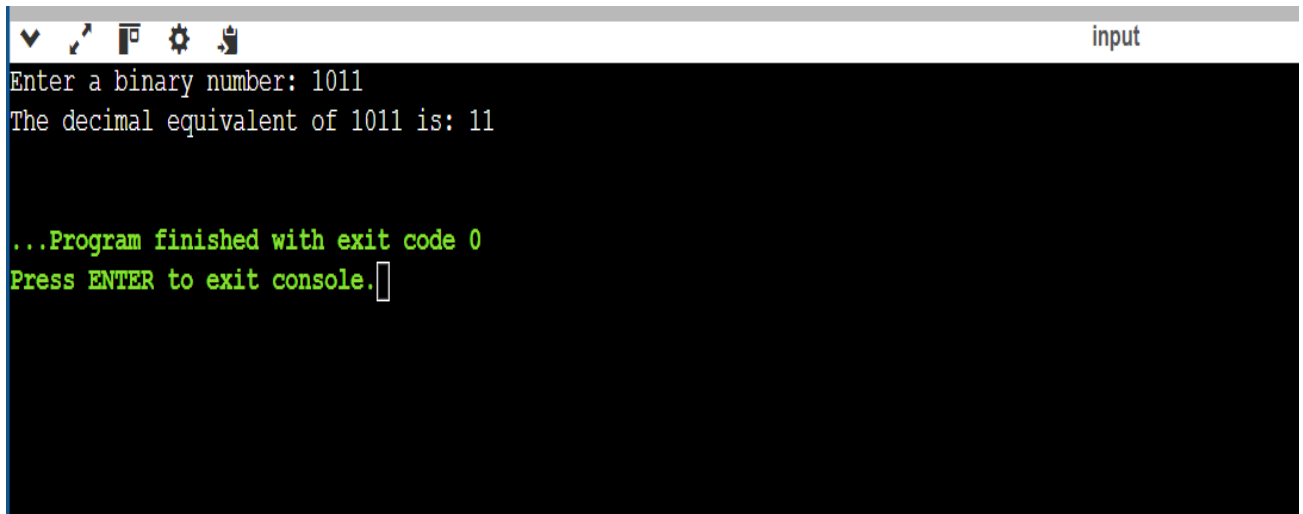
...Program finished with exit code 0
Press ENTER to exit console.
```

- 6) **Write a program to input a binary number in an array and convert it into a corresponding decimal number.**

Input :-

```
#include <stdio.h>
#include <string.h>
#include <math.h>
int binaryToDecimal(char binary[]);
int main() {
    char binary[32];
    int decimal;
    printf("Enter a binary number: ");
    scanf("%s", binary);
    decimal = binaryToDecimal(binary);
    printf("The decimal equivalent of %s is: %d\n", binary, decimal);
    return 0;
}
int binaryToDecimal(char binary[]) {
    int decimal = 0;
    int length = strlen(binary);
    for (int i = 0; i < length; i++) {
        if (binary[i] == '1') {
            decimal += pow(2, length - i - 1);
        }
    }
    return decimal;
}
```

Output :-



```
input
Enter a binary number: 1011
The decimal equivalent of 1011 is: 11

...Program finished with exit code 0
Press ENTER to exit console.
```

7) **Write a program to find the smallest and the largest elements in an array.**

Input :-

```
#include <stdio.h>

int main() {

    int n;

    printf("Enter the number of elements in the array: ");

    scanf("%d", &n);

    int arr[n];

    printf("Enter elements of the array:\n");

    for (int i = 0; i < n; i++) {

        scanf("%d", &arr[i]);

    }

    int smallest = arr[0], largest = arr[0];

    for (int i = 1; i < n; i++) {

        if (arr[i] < smallest) {

            smallest = arr[i];

        }

    }

}
```



```

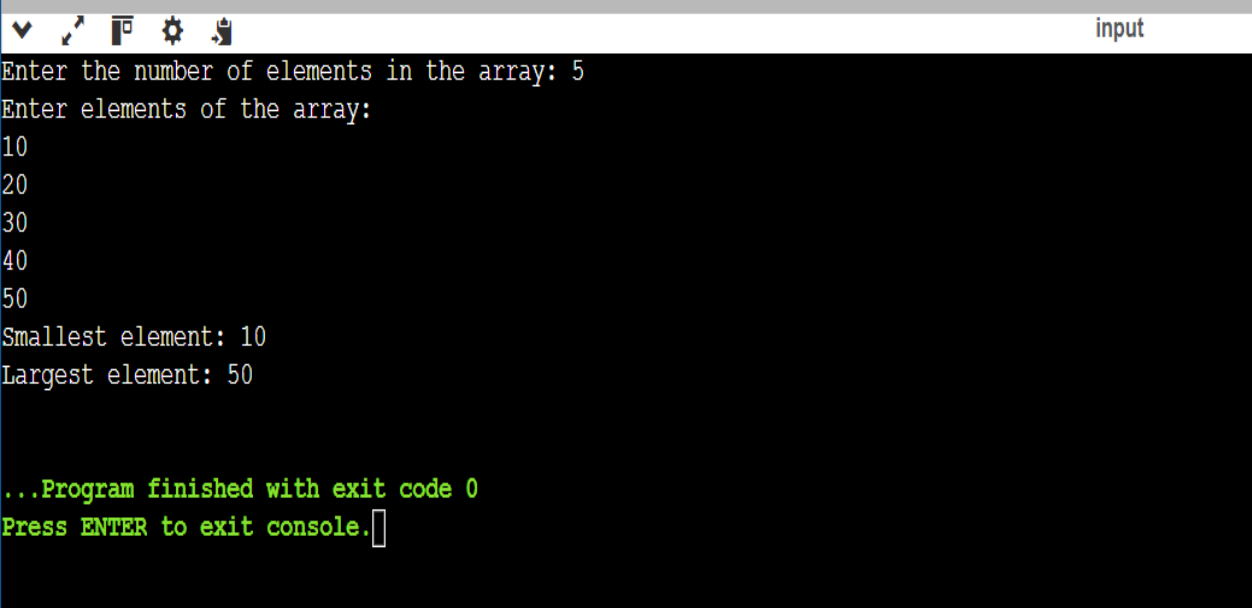
        if (arr[i] > largest) {
            largest = arr[i];
        }
    }

    printf("Smallest element: %d\n", smallest);
    printf("Largest element: %d\n", largest);

    return 0;
}

```

Output :-



```

input
Enter the number of elements in the array: 5
Enter elements of the array:
10
20
30
40
50
Smallest element: 10
Largest element: 50

...Program finished with exit code 0
Press ENTER to exit console.

```

8) Write a program for deleting duplicate elements in an array.

Input :-

```

#include <stdio.h>
int main() {
    int n;
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements of the array:\n");
}

```

```

for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}
int uniqueCount = 0;
for (int i = 0; i < n; i++) {
    int isDuplicate = 0;
    for (int j = 0; j < i; j++) {
        if (arr[i] == arr[j]) {
            isDuplicate = 1;
            break;
        }
    }
    if (!isDuplicate) {
        arr[uniqueCount] = arr[i];
        uniqueCount++;
    }
}
printf("Array without duplicates:\n");
for (int i = 0; i < uniqueCount; i++) {
    printf("%d ", arr[i]);
}
return 0;
}

```

Output :-

```

input
Enter the number of elements in the array: 8
Enter elements of the array:
5
10
8
5
20
20
5
9
Array without duplicates:
5 10 8 20 9
...Program finished with exit code 0
Press ENTER to exit console.

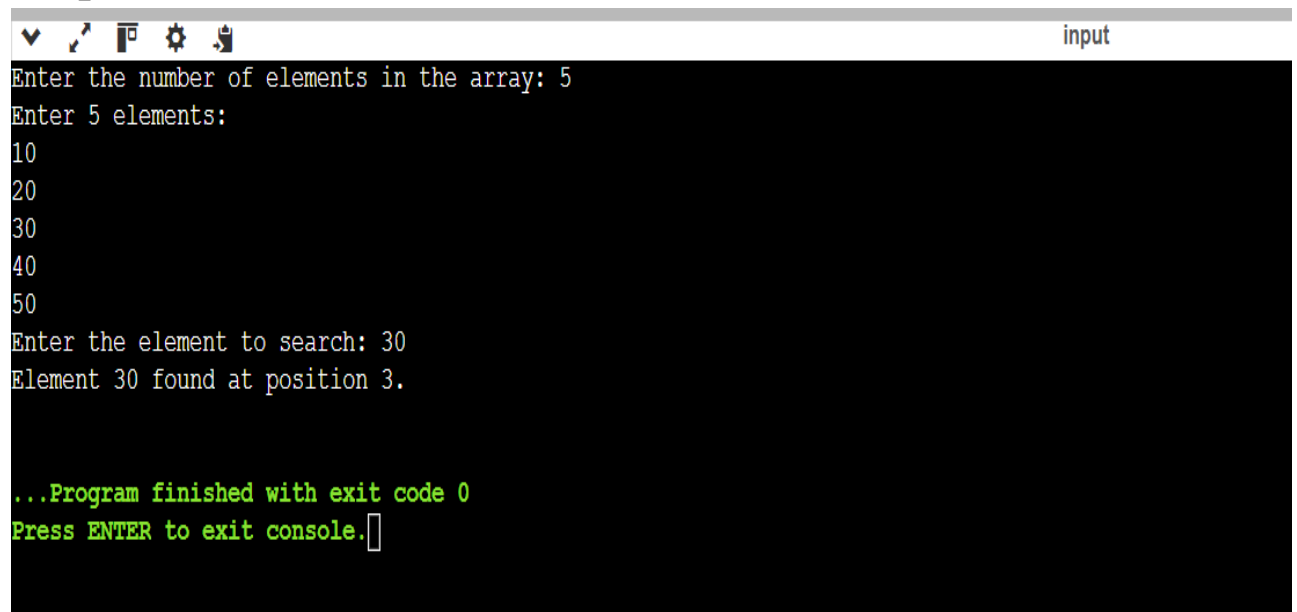
```

- 9) **Write a program to search for a particular element in an array.**

Input :-

```
#include <stdio.h>
int main() {
    int arr[100], n, search, i, found = 0;
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);
    printf("Enter %d elements: \n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the element to search: ");
    scanf("%d", &search);
    for (i = 0; i < n; i++) {
        if (arr[i] == search) {
            printf("Element %d found at position %d.\n", search, i + 1);
            found = 1;
            break;
        }
    }
    if (!found) {
        printf("Element %d not found in the array.\n", search);
    }
    return 0;
}
```

Output :-



```
input
Enter the number of elements in the array: 5
Enter 5 elements:
10
20
30
40
50
Enter the element to search: 30
Element 30 found at position 3.

...Program finished with exit code 0
Press ENTER to exit console.
```

10) Write a program to sort n elements (ascending order).

Input :-

```
#include <stdio.h>

int main() {

    int n;

    printf("Enter the number of elements in the array: ");

    scanf("%d", &n);

    int arr[n];

    printf("Enter elements of the array:\n");

    for (int i = 0; i < n; i++) {

        scanf("%d", &arr[i]);

    }

    for (int i = 0; i < n - 1; i++) {

        for (int j = 0; j < n - i - 1; j++) {

            if (arr[j] > arr[j + 1]) {

                int temp = arr[j];

                arr[j] = arr[j + 1];

                arr[j + 1] = temp;

            }

        }

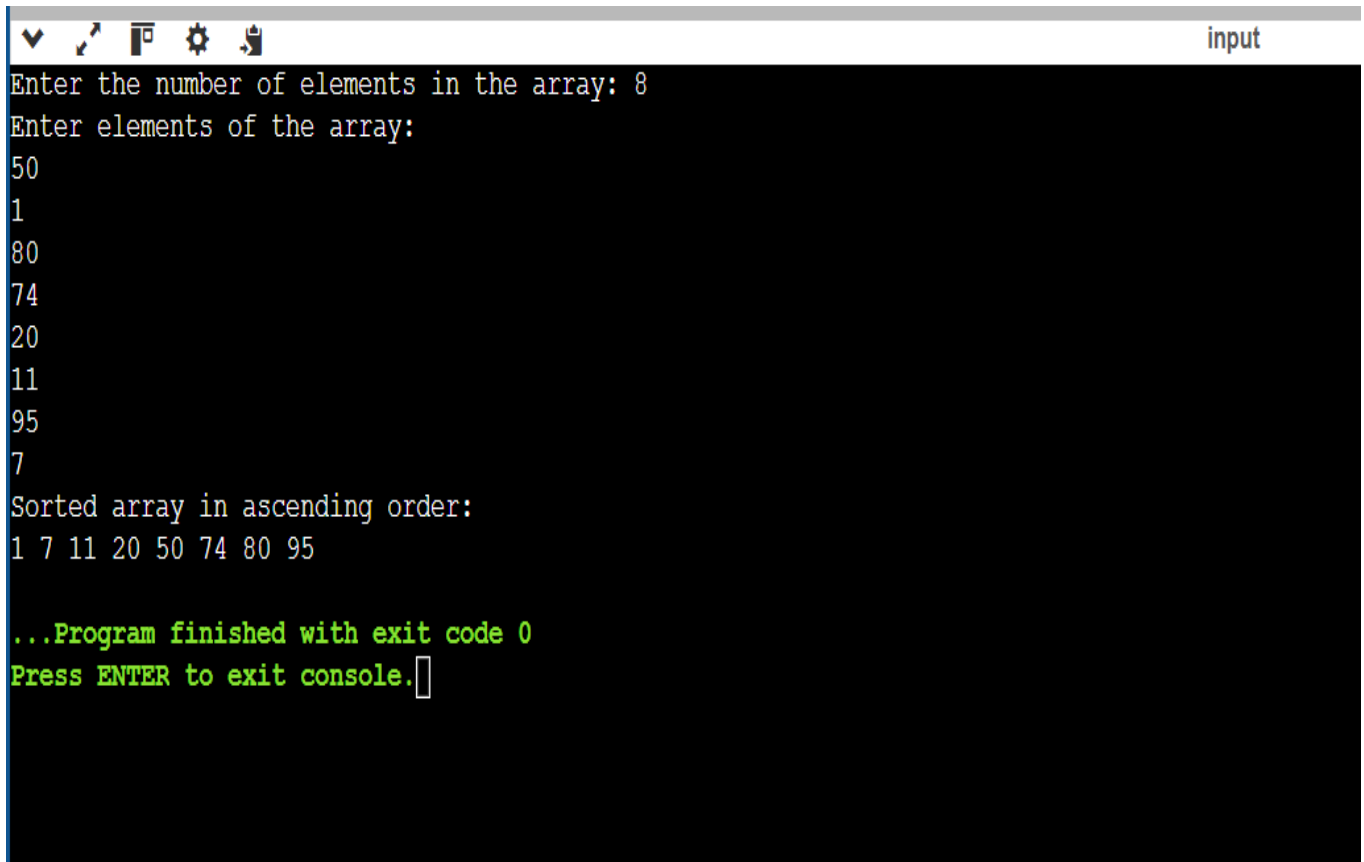
    }

    printf("Sorted array in ascending order:\n");

    for (int i = 0; i < n; i++) {
```

```
    printf("%d ", arr[i]);  
  
}  
  
return 0;  
  
}
```

Output :-



```
input  
Enter the number of elements in the array: 8  
Enter elements of the array:  
50  
1  
80  
74  
20  
11  
95  
7  
Sorted array in ascending order:  
1 7 11 20 50 74 80 95  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

11) Write a program to find the second second-highest number from the array without using sorting.

Input :-

```
#include <stdio.h>  
  
int main() {  
  
    int n;  
  
    printf("Enter the number of elements in the array: ");
```

```
scanf("%d", &n);

int arr[n];

printf("Enter elements of the array:\n");

for (int i = 0; i < n; i++) {

    scanf("%d", &arr[i]);

}

int firstHighest = arr[0];

int secondHighest = arr[0];

for (int i = 1; i < n; i++) {

    if (arr[i] > firstHighest) {

        secondHighest = firstHighest;

        firstHighest = arr[i];

    }

    else if (arr[i] > secondHighest && arr[i] != firstHighest) {

        secondHighest = arr[i];

    }

}

printf("Second highest element: %d\n", secondHighest);

return 0;

}
```

Output :-

```
input
Enter the number of elements in the array: 5
Enter elements of the array:
10
50
80
74
69
Second highest element: 74

...Program finished with exit code 0
Press ENTER to exit console.
```

12) Write a program to perform addition and subtraction between two matrices.

Input :-

```
#include <stdio.h>

int main() {
    int rows, columns, i, j;

    printf("Enter the number of rows and columns of the matrices: ");

    scanf("%d %d", &rows, &columns);

    int matrix1[rows][columns], matrix2[rows][columns], sum[rows][columns],
    diff[rows][columns];

    printf("Enter elements of the first matrix:\n");
```

```

for (i = 0; i < rows; i++) {
    for (j = 0; j < columns; j++) {
        printf("Element [%d][%d]: ", i + 1, j + 1);
        scanf("%d", &matrix1[i][j]);
    }
}

printf("Enter elements of the second matrix:\n");
for (i = 0; i < rows; i++) {
    for (j = 0; j < columns; j++) {
        printf("Element [%d][%d]: ", i + 1, j + 1);
        scanf("%d", &matrix2[i][j]);
    }
}

for (i = 0; i < rows; i++) {
    for (j = 0; j < columns; j++) {
        sum[i][j] = matrix1[i][j] + matrix2[i][j];
        diff[i][j] = matrix1[i][j] - matrix2[i][j];
    }
}

printf("\nSum of the two matrices:\n");
for (i = 0; i < rows; i++) {
    for (j = 0; j < columns; j++) {
        printf("%d\t", sum[i][j]);
    }
}

```



```

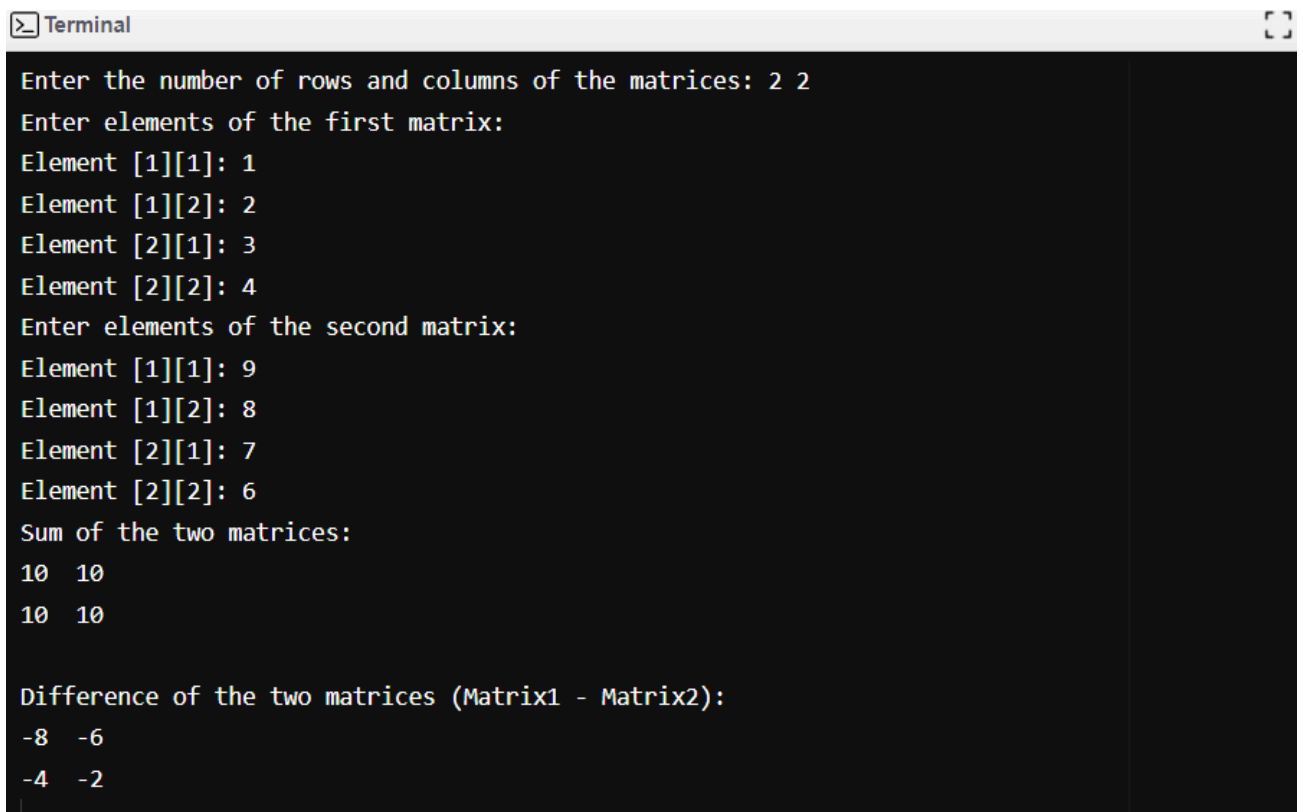
        printf("\n");
    }

    printf("\nDifference of the two matrices (Matrix1 - Matrix2):\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < columns; j++) {
            printf("%d\t", diff[i][j]);
        }
        printf("\n");
    }

    return 0;
}

```

Output :-



A terminal window titled "Terminal" with a dark background and light-colored text. The output of the program is as follows:

```

Enter the number of rows and columns of the matrices: 2 2
Enter elements of the first matrix:
Element [1][1]: 1
Element [1][2]: 2
Element [2][1]: 3
Element [2][2]: 4
Enter elements of the second matrix:
Element [1][1]: 9
Element [1][2]: 8
Element [2][1]: 7
Element [2][2]: 6
Sum of the two matrices:
10 10
10 10

Difference of the two matrices (Matrix1 - Matrix2):
-8 -6
-4 -2

```

13) Write a program to transpose a matrix.

Input :-

```
#include <stdio.h>

int main() {

    int rows, columns, i, j;

    printf("Enter the number of rows and columns of the matrix: ");

    scanf("%d %d", &rows, &columns);

    int matrix[rows][columns], transpose[columns][rows];

    printf("Enter elements of the matrix:\n");

    for (i = 0; i < rows; i++) {

        for (j = 0; j < columns; j++) {

            printf("Element [%d][%d]: ", i + 1, j + 1);

            scanf("%d", &matrix[i][j]);

        }

    }

    for (i = 0; i < rows; i++) {

        for (j = 0; j < columns; j++) {

            transpose[j][i] = matrix[i][j];

        }

    }

    printf("\nOriginal Matrix:\n");

    for (i = 0; i < rows; i++) {

        for (j = 0; j < columns; j++) {

            printf("%d\t", matrix[i][j]);

        }

    }

}
```

```
        printf("\n");
    }

    printf("\nTranspose of the Matrix:\n");
    for (i = 0; i < columns; i++) {
        for (j = 0; j < rows; j++) {
            printf("%d\t", transpose[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output :-

 Terminal

```
Enter the number of rows and columns of the matrix: 2 3
Enter elements of the matrix:
Element [1][1]: 1
Element [1][2]: 2
Element [1][3]: 3
Element [2][1]: 4
Element [2][2]: 5
Element [2][3]: 6
Original Matrix:
1   2   3
4   5   6

Transpose of the Matrix:
1   4
2   5
3   6
```

14) Write a program to add the elements of each row and each column of a matrix.

Input :-

```
#include <stdio.h>

int main() {

    int rows, columns, i, j;

    printf("Enter the number of rows and columns of the matrix: ");

    scanf("%d %d", &rows, &columns);

    int matrix[rows][columns], rowSum[rows], colSum[columns];

    printf("Enter elements of the matrix:\n");

    for (i = 0; i < rows; i++) {

        for (j = 0; j < columns; j++) {

            printf("Element [%d][%d]: ", i + 1, j + 1);

            scanf("%d", &matrix[i][j]);

        }

    }

    for (i = 0; i < rows; i++) {

        rowSum[i] = 0;

    }

    for (j = 0; j < columns; j++) {

        colSum[j] = 0;

    }

    for (i = 0; i < rows; i++) {

        for (j = 0; j < columns; j++) {

            rowSum[i] += matrix[i][j];
```

```

        colSum[j] += matrix[i][j];
    }
}

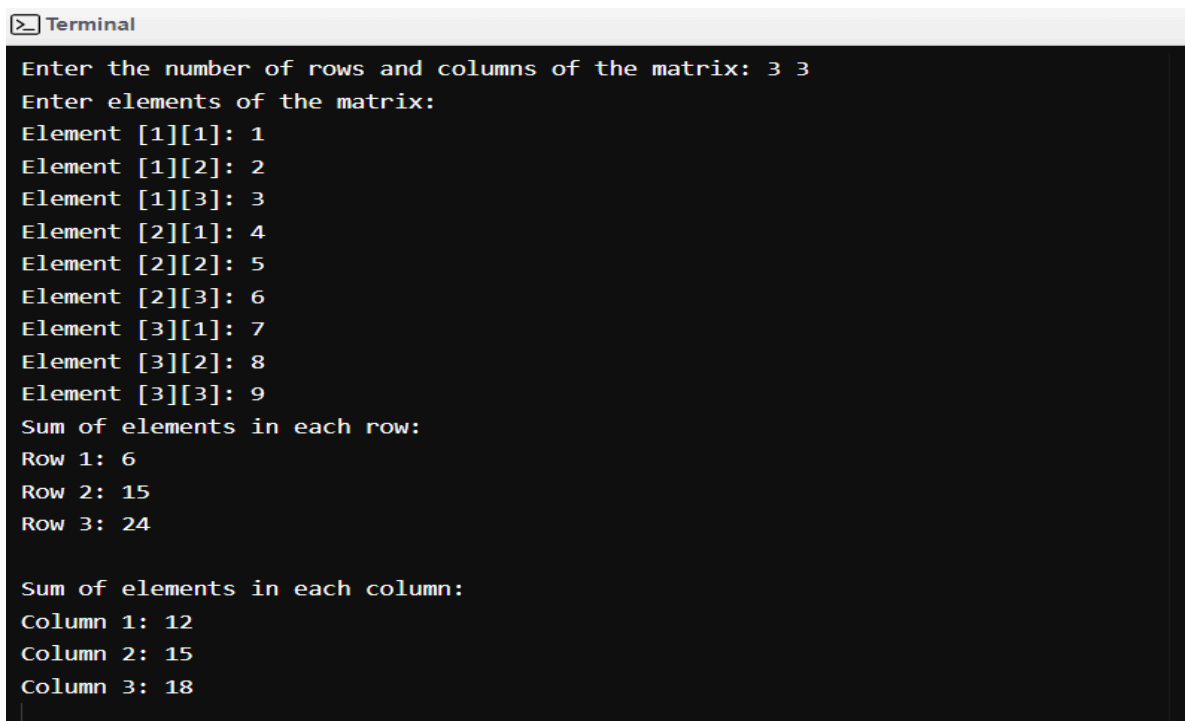
printf("\nSum of elements in each row:\n");
for (i = 0; i < rows; i++) {
    printf("Row %d: %d\n", i + 1, rowSum[i]);
}

printf("\nSum of elements in each column:\n");
for (j = 0; j < columns; j++) {
    printf("Column %d: %d\n", j + 1, colSum[j]);
}

return 0;
}

```

Output :-



```

Terminal
Enter the number of rows and columns of the matrix: 3 3
Enter elements of the matrix:
Element [1][1]: 1
Element [1][2]: 2
Element [1][3]: 3
Element [2][1]: 4
Element [2][2]: 5
Element [2][3]: 6
Element [3][1]: 7
Element [3][2]: 8
Element [3][3]: 9
Sum of elements in each row:
Row 1: 6
Row 2: 15
Row 3: 24

Sum of elements in each column:
Column 1: 12
Column 2: 15
Column 3: 18

```

15) Write a program to perform the multiplication of two matrices.

Input :-

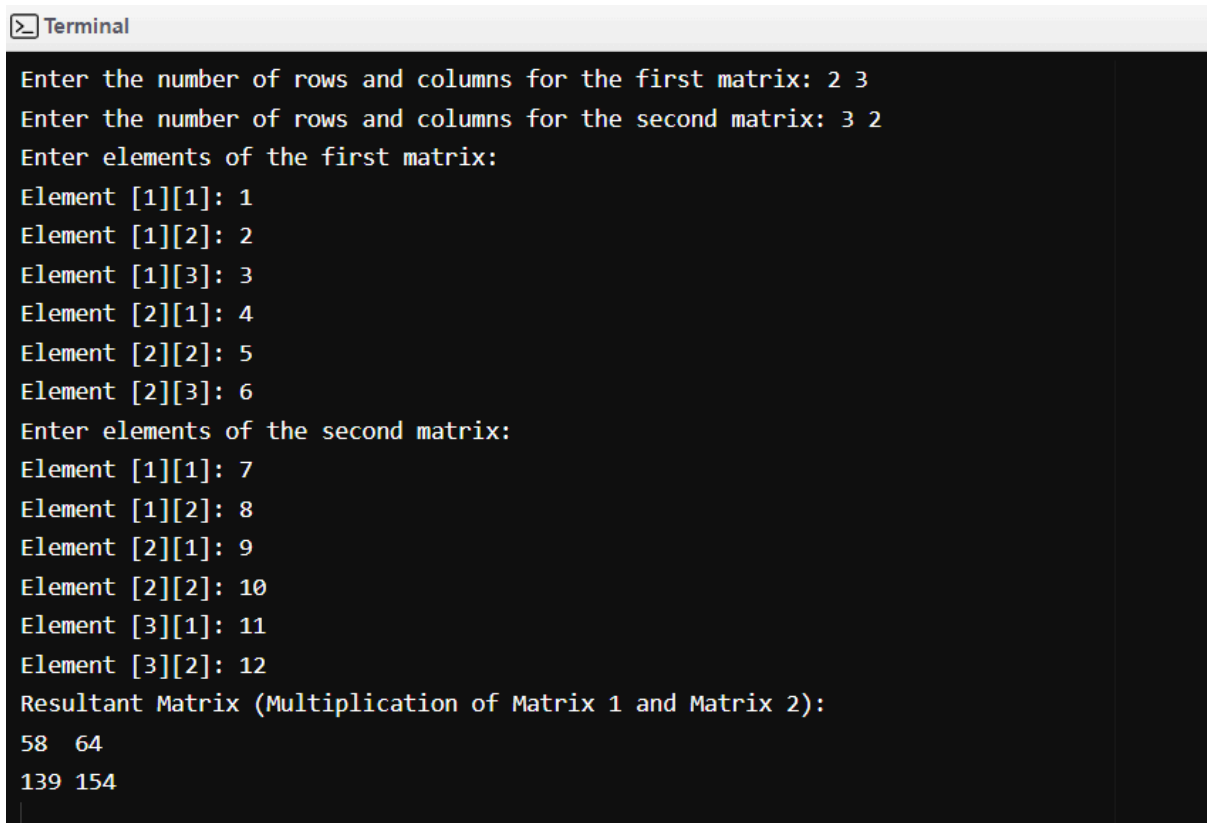
```
#include <stdio.h>
int main() {
    int rows1, cols1, rows2, cols2, i, j, k;
    printf("Enter the number of rows and columns for the first matrix: ");
    scanf("%d %d", &rows1, &cols1);
    printf("Enter the number of rows and columns for the second matrix: ");
    scanf("%d %d", &rows2, &cols2);
    if (cols1 != rows2) {
        printf("Matrix multiplication not possible! The number of columns in the
first matrix must be equal to the number of rows in the second matrix.\n");
        return 0;
    }
    int matrix1[rows1][cols1], matrix2[rows2][cols2], result[rows1][cols2];
    printf("Enter elements of the first matrix:\n");
    for (i = 0; i < rows1; i++) {
        for (j = 0; j < cols1; j++) {
            printf("Element [%d][%d]: ", i + 1, j + 1);
            scanf("%d", &matrix1[i][j]);
        }
    }
    printf("Enter elements of the second matrix:\n");
    for (i = 0; i < rows2; i++) {
        for (j = 0; j < cols2; j++) {
            printf("Element [%d][%d]: ", i + 1, j + 1);
            scanf("%d", &matrix2[i][j]);
        }
    }
    for (i = 0; i < rows1; i++) {
        for (j = 0; j < cols2; j++) {
            result[i][j] = 0;
        }
    }
    for (i = 0; i < rows1; i++) {
        for (j = 0; j < cols2; j++) {
            for (k = 0; k < cols1; k++) {
                result[i][j] += matrix1[i][k] * matrix2[k][j];
            }
        }
    }
}
```

```

    }
}
}
printf("\nResultant Matrix (Multiplication of Matrix 1 and Matrix 2):\n");
for (i = 0; i < rows1; i++) {
    for (j = 0; j < cols2; j++) {
        printf("%d\t", result[i][j]);
    }
    printf("\n");
}
return 0;
}

```

Output :-



```

Terminal
Enter the number of rows and columns for the first matrix: 2 3
Enter the number of rows and columns for the second matrix: 3 2
Enter elements of the first matrix:
Element [1][1]: 1
Element [1][2]: 2
Element [1][3]: 3
Element [2][1]: 4
Element [2][2]: 5
Element [2][3]: 6
Enter elements of the second matrix:
Element [1][1]: 7
Element [1][2]: 8
Element [2][1]: 9
Element [2][2]: 10
Element [3][1]: 11
Element [3][2]: 12
Resultant Matrix (Multiplication of Matrix 1 and Matrix 2):
58  64
139 154

```

16) Write a program to check whether a matrix is an identity matrix or not.

Input :-

```
#include <stdio.h>

int main() {

    int size, i, j, isIdentity = 1;

    printf("Enter the size of the matrix (n for an n x n matrix): ");

    scanf("%d", &size);

    int matrix[size][size];

    printf("Enter elements of the matrix:\n");

    for (i = 0; i < size; i++) {

        for (j = 0; j < size; j++) {

            printf("Element [%d][%d]: ", i + 1, j + 1);

            scanf("%d", &matrix[i][j]);

        }

    }

    for (i = 0; i < size; i++) {

        for (j = 0; j < size; j++) {

            if (i == j) {

                if (matrix[i][j] != 1) {

                    isIdentity = 0;

                    break;

                }

            }

        }

    }

}
```



```
        else {  
            if (matrix[i][j] != 0) {  
                isIdentity = 0;  
                break;  
            }  
        }  
    }  
    if (!isIdentity)  
        break;  
}  
if (isIdentity) {  
    printf("The matrix is an identity matrix.\n");  
}  
else {  
    printf("The matrix is NOT an identity matrix.\n");  
}  
return 0;  
}
```

Output :-

a)

```
Terminal
Enter the size of the matrix (n for an n x n matrix): 3
Enter elements of the matrix:
Element [1][1]: 1
Element [1][2]: 0
Element [1][3]: 0
Element [2][1]: 0
Element [2][2]: 1
Element [2][3]: 0
Element [3][1]: 0
Element [3][2]: 0
Element [3][3]: 1
The matrix is an identity matrix.
```

b)

```
Terminal
Enter the size of the matrix (n for an n x n matrix): 3
Enter elements of the matrix:
Element [1][1]: 1
Element [1][2]: 2
Element [1][3]: 0
Element [2][1]: 0
Element [2][2]: 1
Element [2][3]: 0
Element [3][1]: 0
Element [3][2]: 0
Element [3][3]: 1
The matrix is NOT an identity matrix.
```

17) Write a program to check whether a matrix is a sparse matrix or not

Input :-

```
#include <stdio.h>

#define MAX_ROWS 100

#define MAX_COLS 100

int main() {

    int rows, cols, count = 0;

    int matrix[MAX_ROWS][MAX_COLS];

    printf("Enter the number of rows and columns: ");

    scanf("%d %d", &rows, &cols);

    printf("Enter elements of the matrix:\n");

    for (int i = 0; i < rows; i++) {

        for (int j = 0; j < cols; j++) {

            scanf("%d", &matrix[i][j]);

            if (matrix[i][j] != 0) {

                count++;

            }

        }

    }

    if (count <= (rows * cols) / 2) {

        printf("The matrix is a sparse matrix.\n");

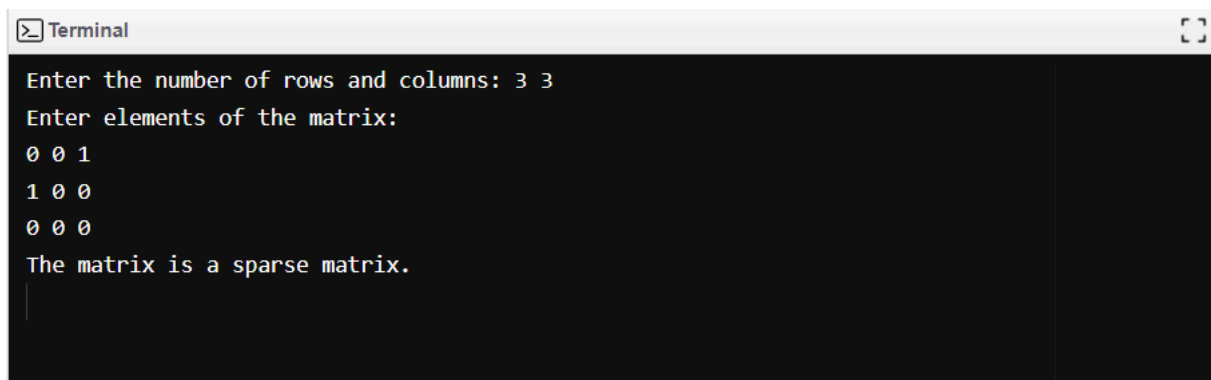
    }

}
```

```
}  
  
else {  
  
    printf("The matrix is not a sparse matrix.\n");  
  
}  
  
return 0;  
  
}
```

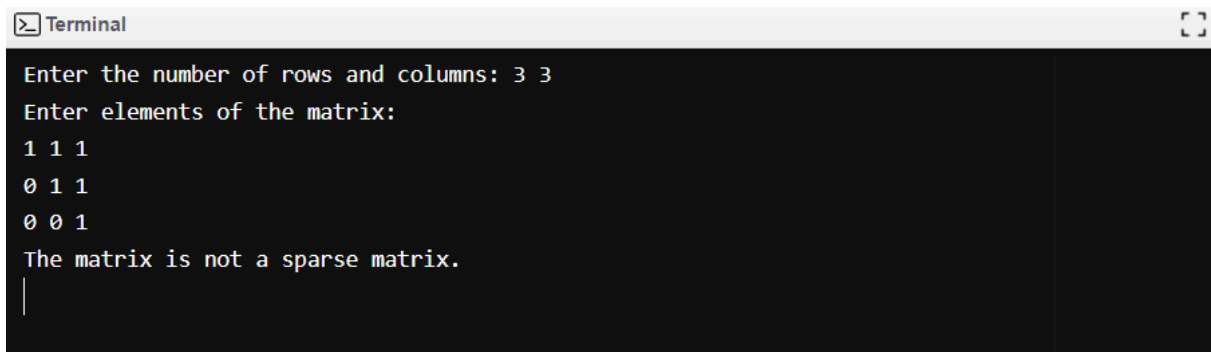
Output :-

a)



```
Terminal  
Enter the number of rows and columns: 3 3  
Enter elements of the matrix:  
0 0 1  
1 0 0  
0 0 0  
The matrix is a sparse matrix.  
|
```

b)



```
Terminal  
Enter the number of rows and columns: 3 3  
Enter elements of the matrix:  
1 1 1  
0 1 1  
0 0 1  
The matrix is not a sparse matrix.  
|
```

18) Write a C program to create a structure named company which has name, address, phone and no Of Employee as member variables. Read the name of the company, its address, phone and no Of Employees. Finally display these members" values.

Input :-

```
#include <stdio.h>

#include <string.h>

struct company {
    char name[100];
    char address[200];
    char phone[20];
    int no_of_employees;
};

int main() {
    struct company company_info;

    printf("Enter the company name: ");
    fgets(company_info.name, 100, stdin);
    company_info.name[strcspn(company_info.name, "\n")] = 0;

    printf("Enter the company address: ");
    fgets(company_info.address, 200, stdin);
    company_info.address[strcspn(company_info.address, "\n")] = 0;

    printf("Enter the company phone number: ");
    fgets(company_info.phone, 20, stdin);
    company_info.phone[strcspn(company_info.phone, "\n")] = 0;

    printf("Enter the number of employees: ");
```

```
scanf("%d", &company_info.no_of_employees);

printf("\nCompany Information:\n");

printf("Name: %s\n", company_info.name);

printf("Address: %s\n", company_info.address);

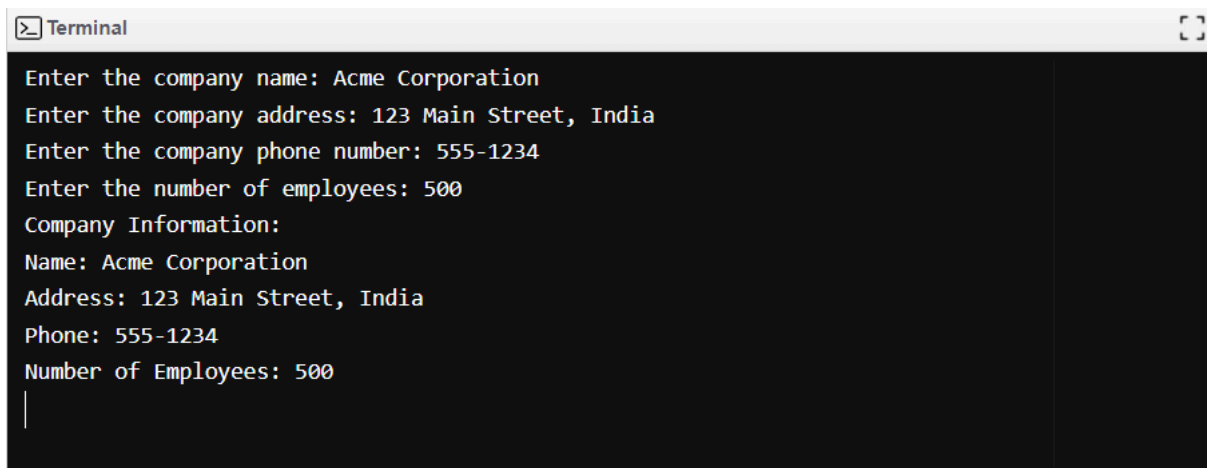
printf("Phone: %s\n", company_info.phone);

printf("Number of Employees: %d\n", company_info.no_of_employees);

return 0;

}
```

Output :-

A screenshot of a terminal window titled "Terminal". The window has a dark background with light-colored text. The text shows the program's execution: it prompts for company name, address, phone number, and number of employees, then displays the collected information in a structured format. The input values are "Acme Corporation", "123 Main Street, India", "555-1234", and "500".

```
Terminal
Enter the company name: Acme Corporation
Enter the company address: 123 Main Street, India
Enter the company phone number: 555-1234
Enter the number of employees: 500
Company Information:
Name: Acme Corporation
Address: 123 Main Street, India
Phone: 555-1234
Number of Employees: 500
|
```

19) Define a structure “complex” (typedef) to read two complex numbers and perform addition, and subtraction of these two complex numbers and display the result.

Input :-

```
#include <stdio.h>

typedef struct {

    float real;

    float imag;
```

```

} complex;

complex add_complex(complex num1, complex num2) {
    complex result;
    result.real = num1.real + num2.real;
    result.imag = num1.imag + num2.imag;
    return result;
}

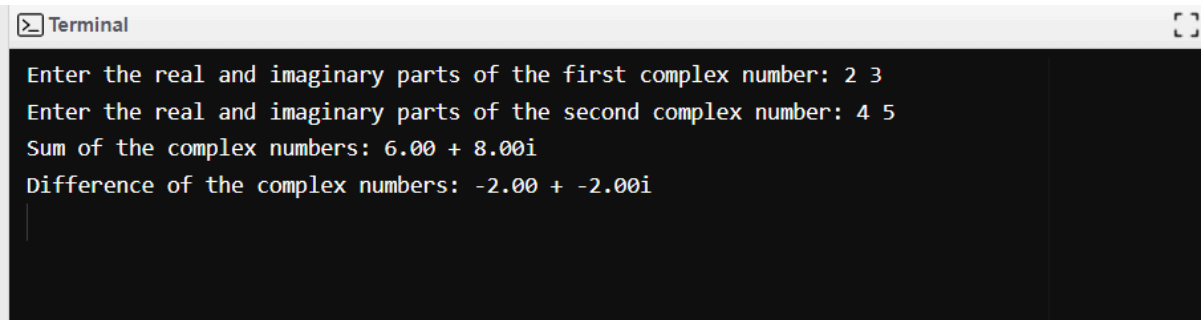
complex subtract_complex(complex num1, complex num2) {
    complex result;
    result.real = num1.real - num2.real;
    result.imag = num1.imag - num2.imag;
    return result;
}

int main() {
    complex num1, num2, sum, difference;
    printf("Enter the real and imaginary parts of the first complex number: ");
    scanf("%f %f", &num1.real, &num1.imag);
    printf("Enter the real and imaginary parts of the second complex number: ");
    scanf("%f %f", &num2.real, &num2.imag);
    sum = add_complex(num1, num2);
    difference = subtract_complex(num1, num2);
    printf("Sum of the complex numbers: %.2f + %.2fi\n", sum.real, sum.imag);
    printf("Difference of the complex numbers: %.2f + %.2fi\n", difference.real,
difference.imag);
    return 0;
}

```

}

Output :-

A terminal window with a title bar that says "Terminal". The window has a dark background with light-colored text. The text inside the terminal shows the execution of a C program. It prompts for the real and imaginary parts of two complex numbers. The first number is 2 + 3i and the second is 4 + 5i. The program then calculates and displays the sum (6.00 + 8.00i) and the difference (-2.00 + -2.00i) of these two complex numbers.

```
Terminal
Enter the real and imaginary parts of the first complex number: 2 3
Enter the real and imaginary parts of the second complex number: 4 5
Sum of the complex numbers: 6.00 + 8.00i
Difference of the complex numbers: -2.00 + -2.00i
```

20) Write a C program to read the RollNo, Name, Address, and Age marks of 12 students in the BCT class and display the details from the function.

Input :-

```
#include <stdio.h>

#include <string.h>

struct student {

    int rollno;

    char name[100];

    char address[200];

    int age;

    float marks;

};

void read_students(struct student students[], int n) {

    for (int i = 0; i < n; i++) {

        printf("Enter details for student %d:\n", i + 1);
```



```

printf("Roll No: ");

scanf("%d", &students[i].rollno);

getchar();

printf("Name: ");

fgets(students[i].name, 100, stdin);

students[i].name[strcspn(students[i].name, "\n")] = '\0';

printf("Address: ");

fgets(students[i].address, 200, stdin);

students[i].address[strcspn(students[i].address, "\n")] = '\0';

printf("Age: ");

scanf("%d", &students[i].age);

printf("Marks: ");

scanf("%f", &students[i].marks);

}

}

void display_students(struct student students[], int n) {

printf("\nStudent Details:\n");

printf("-----\n");

printf("Roll No\tName\tAddress\tAge\tMarks\n");

printf("-----\n");

for (int i = 0; i < n; i++) {

printf("%d\t%-15s\t%-20s\t%d\t%.2f\n", students[i].rollno,
students[i].name, students[i].address, students[i].age, students[i].marks);

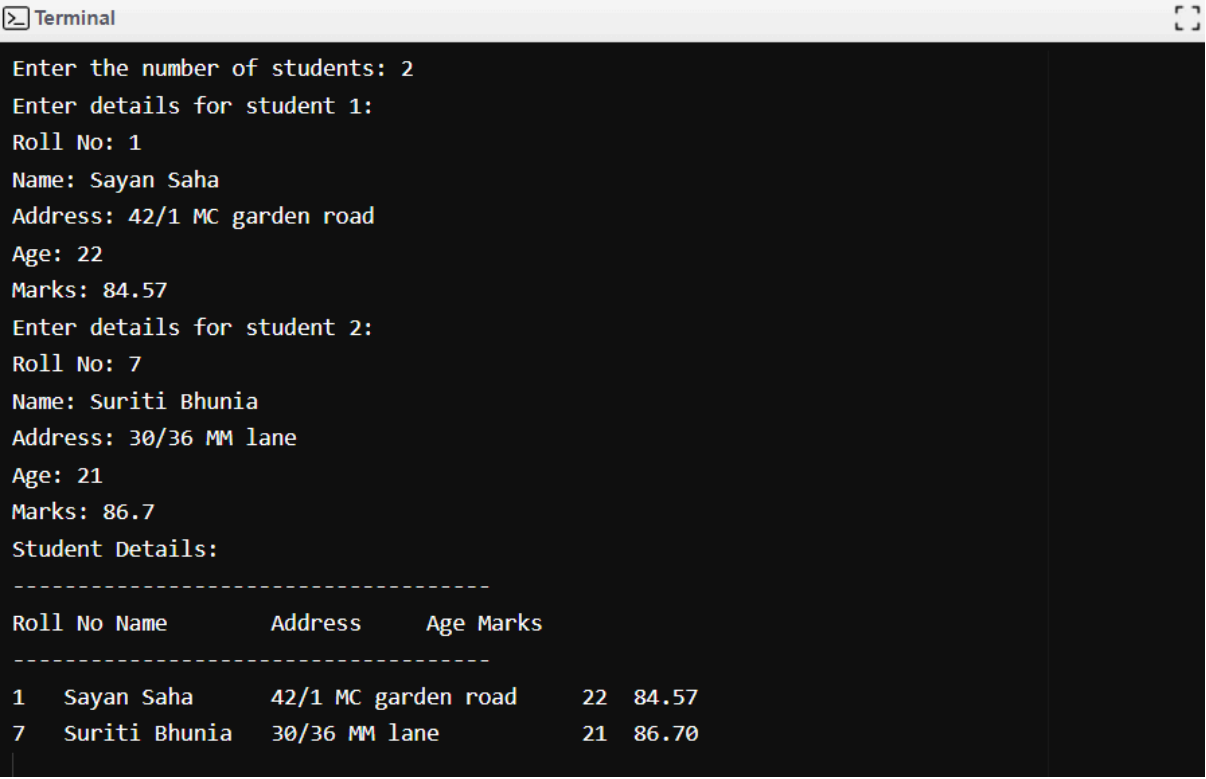
}

}

```

```
int main() {  
    int n;  
  
    printf("Enter the number of students: ");  
  
    scanf("%d", &n);  
  
    struct student students[n];  
  
    read_students(students, n);  
  
    display_students(students, n);  
  
    return 0;  
}
```

Output :-

A terminal window titled "Terminal" with a dark background and light text. It shows the execution of a C program. The user enters '2' for the number of students. Then, for each student, they enter details: Student 1 (Roll No: 1, Name: Sayan Saha, Address: 42/1 MC garden road, Age: 22, Marks: 84.57) and Student 2 (Roll No: 7, Name: Suriti Bhunia, Address: 30/36 MM lane, Age: 21, Marks: 86.7). Finally, the program displays a table of student details.

```
Terminal  
Enter the number of students: 2  
Enter details for student 1:  
Roll No: 1  
Name: Sayan Saha  
Address: 42/1 MC garden road  
Age: 22  
Marks: 84.57  
Enter details for student 2:  
Roll No: 7  
Name: Suriti Bhunia  
Address: 30/36 MM lane  
Age: 21  
Marks: 86.7  
Student Details:  
-----  
Roll No Name      Address      Age Marks  
-----  
1   Sayan Saha    42/1 MC garden road    22 84.57  
7   Suriti Bhunia 30/36 MM lane          21 86.70
```

